STM32WLE5 LoRa Sensor Node

Condensed hardware design report | Mixed-signal sensing + Sub-GHz LoRa + 4-layer PCB

STM32WLE5CC	14-bit ADC	BME280 + IMU	4-layer PCB
MCU + integrated Sub-GHz radio	External precision acquisition	Environmental + motion sensing	50 ohm RF + split power domains

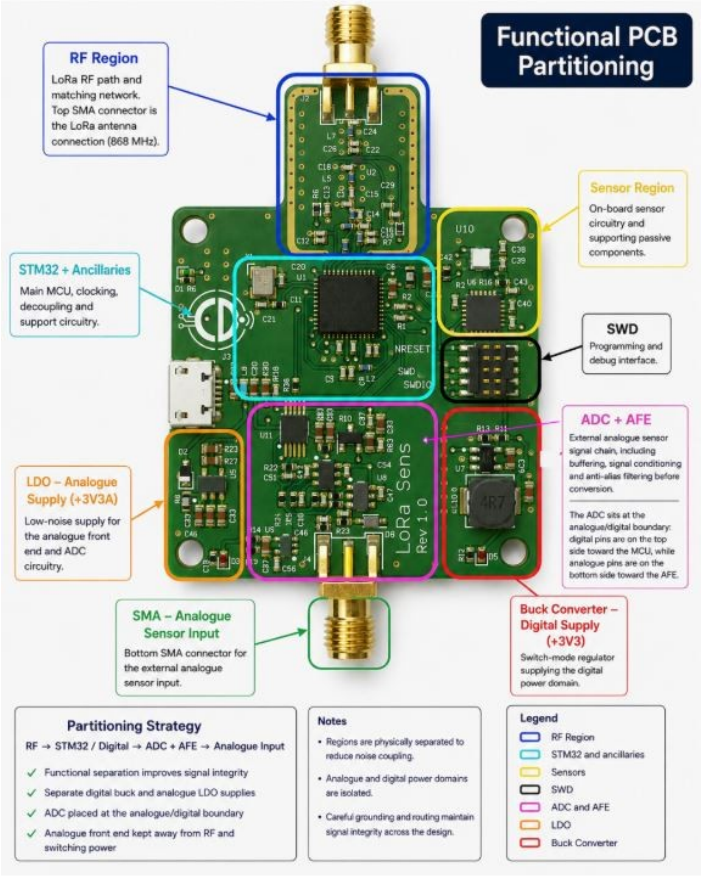


Figure 1 - Functional partitioning of the finished PCB.

Project summary

I designed this board to combine the main electronics disciplines needed in a compact wireless sensor node: signal acquisition, embedded processing, Sub-GHz RF, power architecture and a manufacturable mixed-signal PCB.

What the board demonstrates

- STM32WLE5 MCU with integrated Sub-GHz / LoRa radio.
- Onboard IMU and environmental sensing over I2C.
- External analog sensor input with protection, buffering, anti-alias filtering and a dedicated 14-bit ADC.
- Separate efficient digital SMPS and low-noise analog LDO supply domains.
- Deliberate RF / digital / analog partitioning on a 4-layer PCB.
- 50 ohm controlled-impedance RF path, SWD/debug and production-oriented interfaces.

Design intent at a glance

The partitioning view shows the engineering intent at a glance: the RF section is isolated at the antenna end, the MCU and sensors sit in the digital region, the ADC bridges the analog/digital boundary, and the analog front end is kept away from RF and switching power.

1. System Architecture and PCB Implementation

The schematic architecture and physical layout follow the same signal flow and functional boundaries.

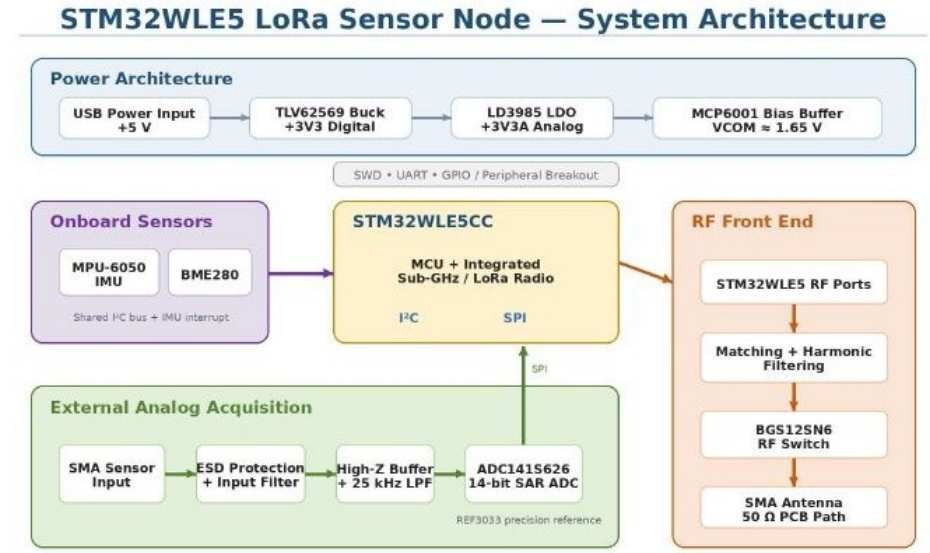


Figure 2 - System architecture: power, onboard sensors, external analog acquisition and STM32WLE5 Sub-GHz RF front end.

Design intent

- Keep the sensor, analog, digital and RF functions distinct from the start rather than resolving conflicts late in layout.
- Use the STM32WL to combine MCU and Sub-GHz radio functions in one device and reduce system complexity.
- Give the analog path its own low-noise supply and keep it physically separated from the digital buck regulator and RF section.
- Place the ADC at the natural boundary between the analog front end and the MCU SPI interface.
- Design the RF path around short routes, local grounding and controlled impedance.

Routed PCB evidence

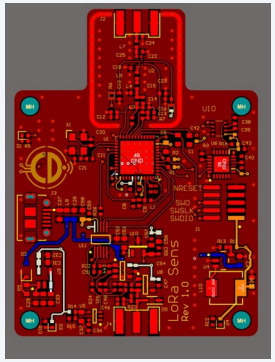


Figure 3 - Routed 4-layer PCB showing the architecture translated into physical placement and routing.

2. MCU, Onboard Sensors and Firmware-Facing Interfaces

A digital sensing path for motion/environmental data, with interrupt-driven operation available for low-power use.

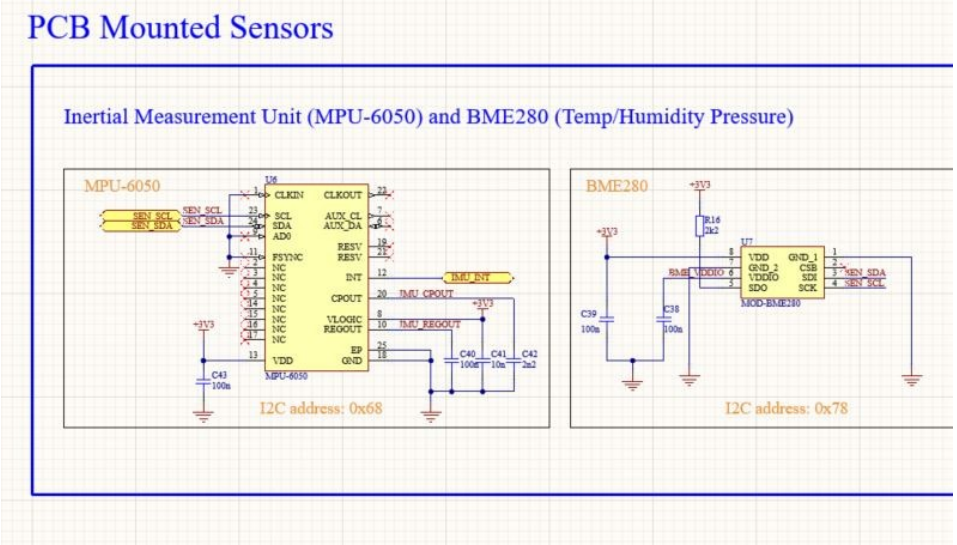


Figure 4 - BME280 environmental sensor and MPU-6050 IMU on a shared I2C bus.

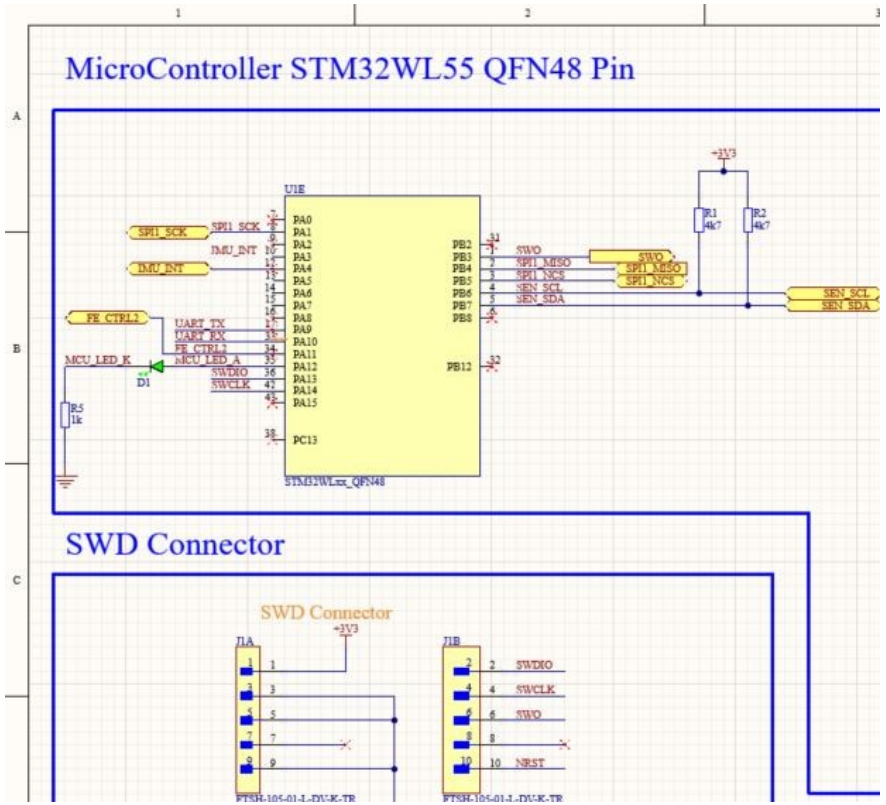


Figure 5 - STM32WLE5 MCU connections with SWD debug access and sensor/peripheral interfaces.

Onboard sensing

- BME280: temperature, humidity and pressure.
- MPU-6050: 3-axis acceleration + 3-axis angular rate.
- Shared I2C bus keeps MCU pin use low.

Low-power behaviour

- IMU interrupt is routed to the MCU.
- Allows event-driven wake-up rather than continuous polling.
- Supports a duty-cycled sensing strategy at firmware level.

Engineering interfaces

- SWD for programming and hardware debug.
- SPI reserved for the external ADC path.
- I2C/UART/GPIO breakout supports expansion and bring-up.

3. External Analog Acquisition Path

Protected external sensor input -> high-impedance buffer -> anti-alias filtering -> dedicated 14-bit ADC.

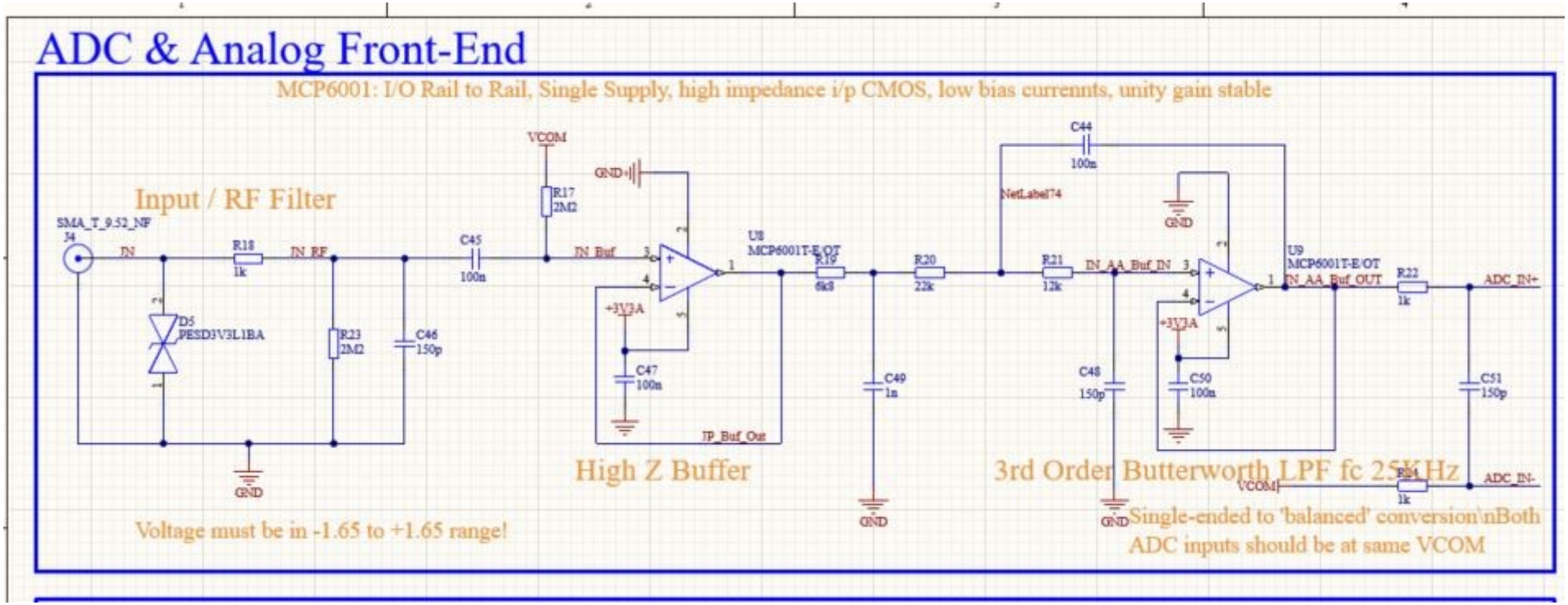


Figure 6 - External analog input and AFE: protection, high-Z buffer and third-order low-pass filtering before the ADC.

Input protection	High-Z buffer	Anti-alias filter	ADC + reference
The SMA sensor input includes transient/ESD protection and an initial RF/noise filter before the active analog stages.	I used an MCP6001 stage so the external sensor is not unnecessarily loaded and the following filter sees a predictable source impedance.	A first-order RC stage and an active second-order Sallen-Key stage form a third-order low-pass response around 25 kHz, limiting out-of-band energy before sampling.	A dedicated ADC141S626 14-bit converter, REF3033 reference and VCOM bias keep conversion separate from the MCU digital rail.

Application flexibility: The external analogue acquisition chain demonstrates protection, buffering, anti-alias filtering and dedicated ADC conversion. For a specific sensor application, the bandwidth, input range and sensor interface would be tailored to the required signal and dynamic range.

4. Sub-GHz RF Design and 4-Layer Controlled-Impedance PCB

STM32WL RF output, matching/filtering, RF switch and a controlled 50 ohm route to the SMA antenna connector.

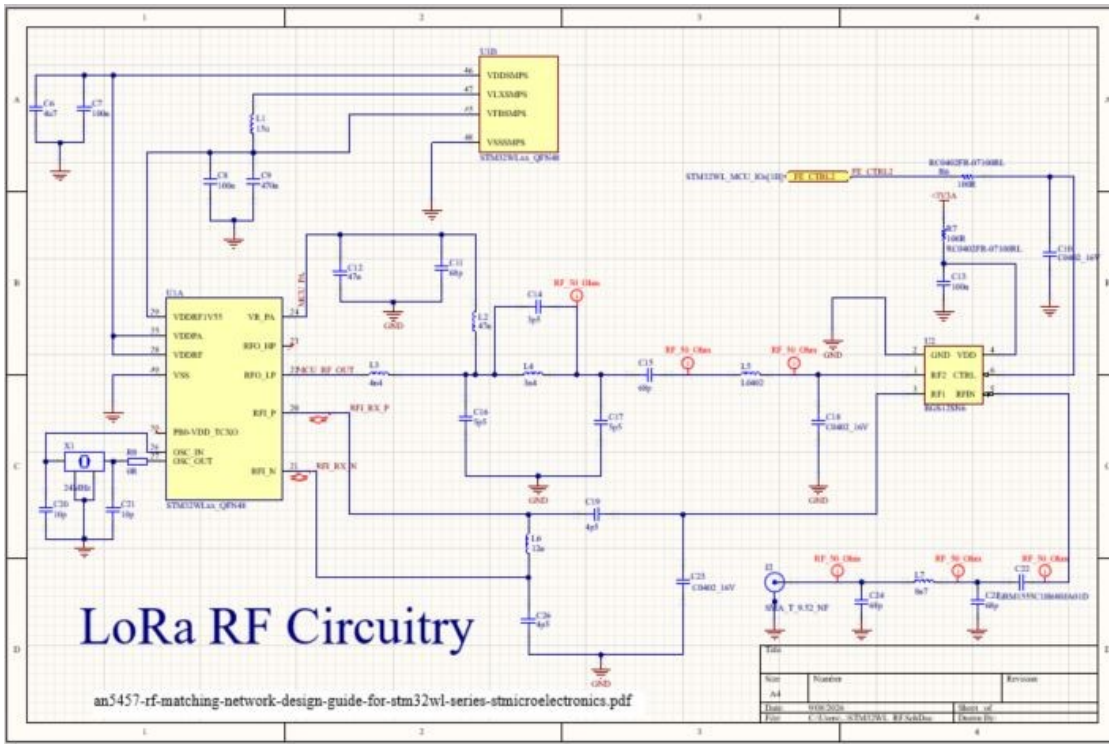


Figure 7 - LoRa/Sub-GHz RF circuitry: STM32WL RF ports, matching/filter network, BGS12SN6 RF switch and antenna path.

RF design intent

- Integrated STM32WL Sub-GHz radio reduces the need for a separate MCU + transceiver.
- Matching and harmonic filtering were treated as a dedicated RF signal chain rather than ordinary digital routing.
- The RF section is physically isolated at the top of the PCB with short routes and local grounding.
- The antenna connection is carried on a 50 ohm controlled-impedance coplanar route.

Layer stack / impedance evidence

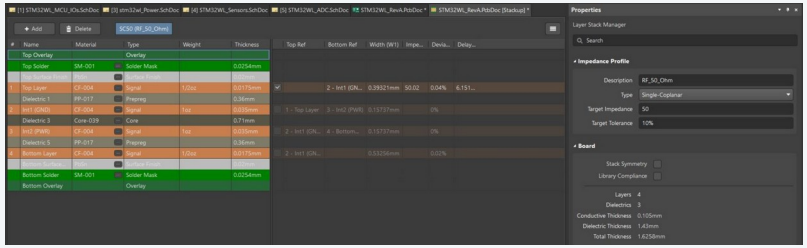
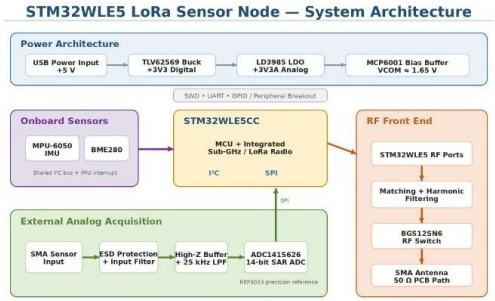


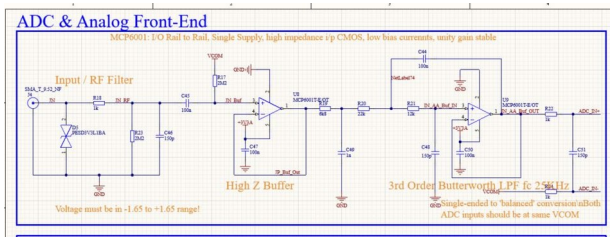
Figure 8 - Altium layer-stack manager showing the 4-layer construction and 50 ohm single-coplanar impedance profile.

5. Engineering Evidence and Application Development Considerations

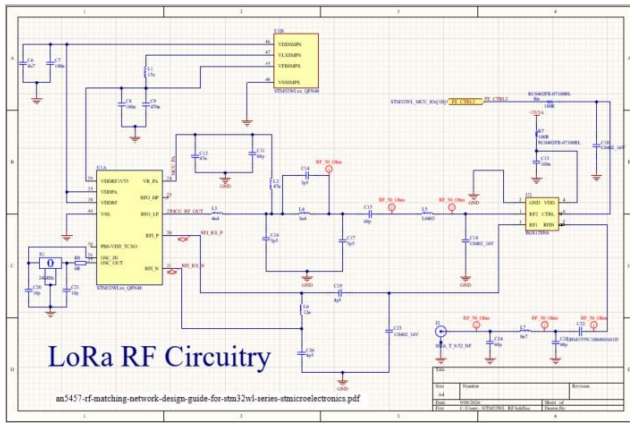
What this project demonstrates today - and what would still be validated on a field product.



System architecture



Analog acquisition chain



Sub-GHz RF signal chain

Evidence demonstrated by this design

- STM32WLE5 integrated MCU + Sub-GHz/LoRa hardware architecture.
- Onboard motion sensing with an interrupt path suitable for event-driven wake-up.
- External analog sensor input with protection, conditioning, anti-alias filtering and dedicated ADC conversion.
- Separate digital SMPS and low-noise analog LDO supply domains.
- 4-layer mixed-signal PCB with explicit RF/digital/analog partitioning and 50 ohm RF routing.
- SWD/debug, I2C, SPI and breakout interfaces designed for bring-up and firmware integration.

Application-specific adaptation and verification

- Select sensors to meet the intended measurement range, bandwidth, resolution and noise requirements.
- Adapt analogue filtering, ADC sampling and processing to the target sensor and signal characteristics.
- Define the power source and measure sleep, acquisition and transmit current before stating battery life.
- Develop and verify application firmware, wireless reliability, security and any required update mechanism.
- Confirm RF link performance using link-budget analysis, measurements and staged field testing.
- Complete environmental, enclosure and long-duration verification for the intended deployment.

Evidence boundary: This report documents schematic, simulation and PCB design evidence. It does not claim completed physical bring-up, measured RF range, demonstrated battery lifetime or validated production firmware. Those require application-specific hardware testing.

6. Supporting Simulation and Analysis Evidence

Selected filter and RF plots used to verify the design intent before hardware manufacture.

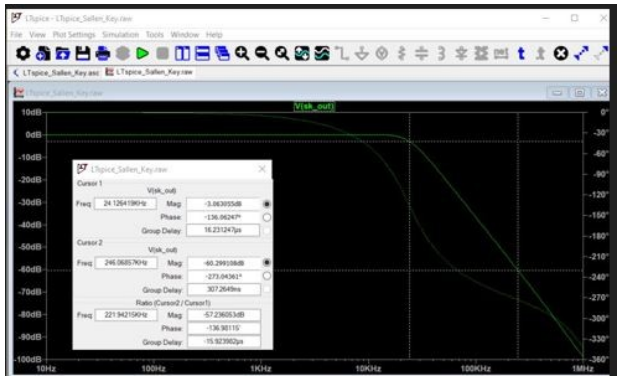


Figure 9 - LTspice bode plot of the anti-alias filter, showing an approximate 25 kHz corner.

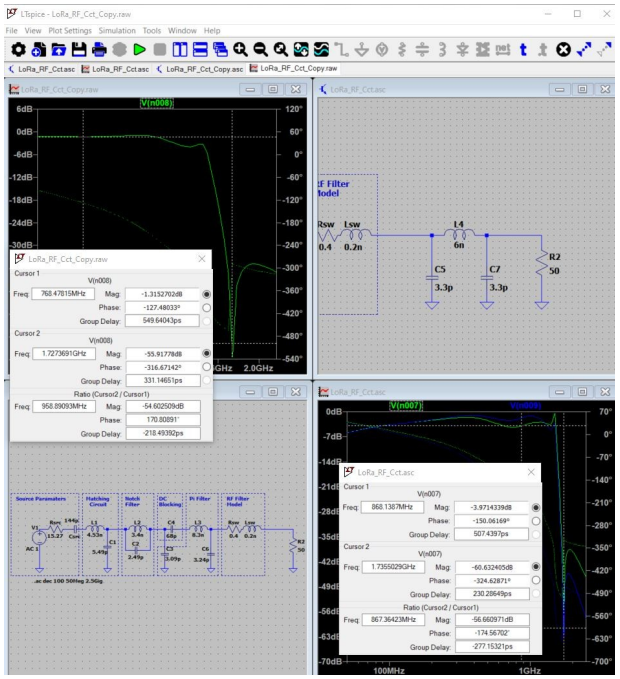


Figure 10 - RF matching/filter simulation around the 868 MHz operating band.

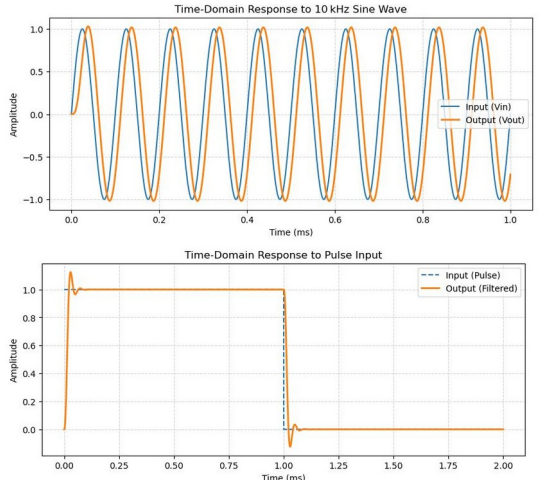


Figure 11 - Time-domain response confirming the filtered sine-wave and transient behaviour.

Simulation-Based Verification Summary

- The bode plot confirms that the external anti-alias stage is tuned to the intended ~25 kHz region with the expected roll-off.
- The time-domain result checks that the analog path preserves the wanted waveform while maintaining a controlled transient response.
- The RF simulation provides direct evidence that the Sub-GHz matching and filtering were treated as an engineered signal chain around 868 MHz.
- Together, these results provide simulation-based design verification; final measurements on assembled hardware remain necessary.

Taken together, these analyses support the analog filtering and Sub-GHz RF design choices at simulation stage. Bench measurements on assembled hardware would be the next verification step.